

Exercise 1: simple perceptron

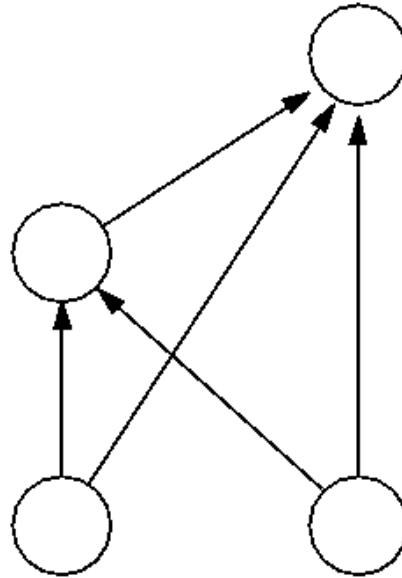
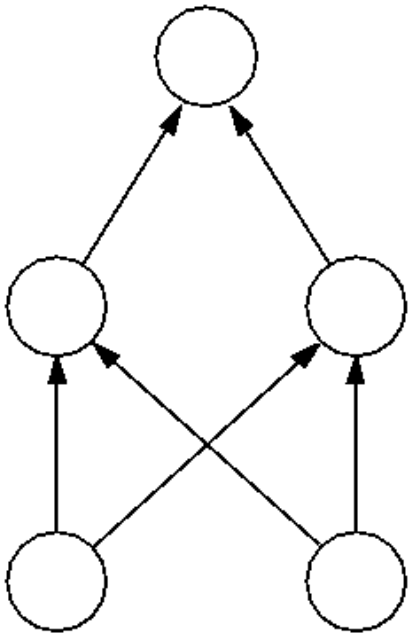
(1) Draw the input space for the following classification:

x_1	x_2	target
0	0	0
0.5	0	0
0	1	1
1.5	0	1

(2) Draw a possible decision line in input space. What is the equation describing this line, and what are possible weights and a threshold that would result in such a decision line?

Exercise 2: Using a two-layer network to solve XOR

- (1) Find possible weights and thresholds that solve the XOR problem.
- (2) Draw the decision lines in input space.



x_1	x_2	$y = x_1 \text{ XOR } x_2$
0	0	0
0	1	1
1	0	1
1	1	0